

Electromagnetic Compatibility: BONDING



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In the previous parts of this article, we read about high-speed PCB design, multi-layer PCB design and grounding for electromagnetic compatibility. Let us now look at electrical bonding.

Electrical bonding is a process by which parts in an equipment assembly or sub-systems within a system are connected electrically by their joints or by any low-resistance bonding media. The main purpose of a bond is to make a structure homogenous with respect to the flow of radio frequency (RF) current, so that it would experience minimum barrier as it crosses one surface to the other without developing electrical potential at the crossover point.

To ensure electromagnetic compatibility (EMC) in an equipment or a system, the various components, modules, cable shields, connectors, etc that make up the system need to be connected to a chassis or reference ground via a low-impedance bond that should provide near-zero impedance at all frequencies, or at least over those frequencies for which EMC standards require measurements to be carried out.

Good bonding is required, for example, for mounting line-filter modules on the chassis that serves as a drain for EMI currents, or for connector shells to equipment enclosures to ensure shielding integrity of cable shields that are terminated on these connectors, or for ensuring shielding

integrity over seams and joints to avoid leakage of RF energy.

Behaviour of a bond at RFs

At direct current (DC) or power frequencies, bond resistance should be less than one milli-ohm. Even if this is ensured, behavior of the bond at RF can be very different, since parasitic effects set in and the bond is no longer purely resistive. Fig. 21 represents the bond at RF where you can see that, in addition to resistance, there is now self-inductance in series and residual capacitance in parallel.

This now becomes a parallel resonant circuit and at its resonant frequency (f_r) it exhibits a high impedance. In addition, at higher frequencies, due to skin effect, current tends to flow via the outer periphery of the conductor, and the bond offers a high resistance.

Types of bonding

The intention of bonding is to prevent a voltage difference between the members being joined. To perform effectively at DC or low frequencies, including 50Hz or 60Hz power frequencies, a simple (but durable and permanent) low-resistance joint is adequate.

To accomplish a low DC resistance bond between two adjoining cabinets, bolt holes can be drilled in the adjoining parallel cabinet walls and the two cabinets can be bolted together with star washers at the points of contact between the cabinet walls.

But such a bond is deficient at high frequencies because of higher resistance and inductive reactance. This is more marked where the bond involves dissimilar metals where non-linear junctions can develop at the bonds, leading to the generation of spurious and harmonics, especially under the presence of strong

Fig. 21: Bond at high frequency

